AIM:

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:-

```
  GNU nano 8.7                                              Optimal.c
#include <stdio.h>

int main()
{
    int frames[10], pages[50];
    int n, f, i, j, k, pos, max, faults = 0, hits = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i = 0; i < f; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++)
    {
        int found = 0;

        for(j = 0; j < f; j++)
        {
            if(frames[j] == pages[i])
            {
                found = 1;
                hits++;

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Exec
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Just
```

```
  GNU nano 8.7                                              Optimal.c
                found = 1;
                hits++;
                break;
            }
        }

        if(found == 0)
        {
            pos = -1;
            max = -1;

            for(j = 0; j < f; j++)
            {
                int next = -1;

                for(k = i + 1; k < n; k++)
                {
                    if(frames[j] == pages[k])
                    {
                        next = k;
                        break;
                    }
                }

                if(next == -1)
                {
                    pos = j;
                    break;
                }

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify
```

```
  GNU nano 8.7                                              Optimal.c
                    pos = j;
                    break;
                }

                if(next > max)
                {
                    max = next;
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            faults++;
        }

        printf("Frames: ");
        for(j = 0; j < f; j++)
            printf("%d ", frames[j]);
        printf("\n");
    }

    printf("\nTotal Page Faults = %d", faults);
    printf("\nTotal Page Hits = %d", hits);

    printf("\nPage Fault Percentage = %.2f%%", (float)faults/n*100);
    printf("\nPage Hit Percentage = %.2f%%", (float)hits/n*100);

    return 0;
}

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify
```

OUTPUT:-

```
Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ nano Optimal.c

Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ gcc Optimal.c -o Optimal

Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ ./Optimal
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3
Frames: 1 -1 -1
Frames: 1 2 -1
Frames: 1 2 3
Frames: 1 2 4
Frames: 1 2 4
Frames: 1 2 4
Frames: 1 2 5
Frames: 1 2 5
Frames: 1 2 5
Frames: 3 2 5
Frames: 4 2 5
Frames: 4 2 5

Total Page Faults = 7
Total Page Hits = 5
Page Fault Percentage = 58.33%
Page Hit Percentage = 41.67%
Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ |
```